

Shrinkage Estimation of Covariance Matrices: Economic Loss Minimization with Reinforcement Learning

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KEY TAKEAWAY

By learning a regime-dependent shrinkage intensity that directly optimizes an economic loss instead of a statistical surrogate, the proposed covariance matrix estimator recovers one-third of the gap between policy learning and the oracle.

MOTIVATION

- ★ **Covariance shrinkage** is fundamental for portfolio optimization, return prediction, and risk management.
- ★ Goal: minimize the **economic loss (future portfolio risk)**, not the statistical one.

ECONOMIC IDEA

$$\hat{\rho}_{ij}(a) = (1-a) \rho_{ij} \cdot \frac{\sigma_i \sigma_j}{\hat{\sigma}_i \hat{\sigma}_j} \xrightarrow{a \rightarrow 1} 0 \quad (i \neq j)$$
$$\hat{\sigma}_i^2(a) = (1-a) \sigma_i^2 + a \bar{\sigma}^2 \xrightarrow{a \rightarrow 1} \bar{\sigma}^2$$

- ★ Optimal shrinkage → **regime-dependent**.

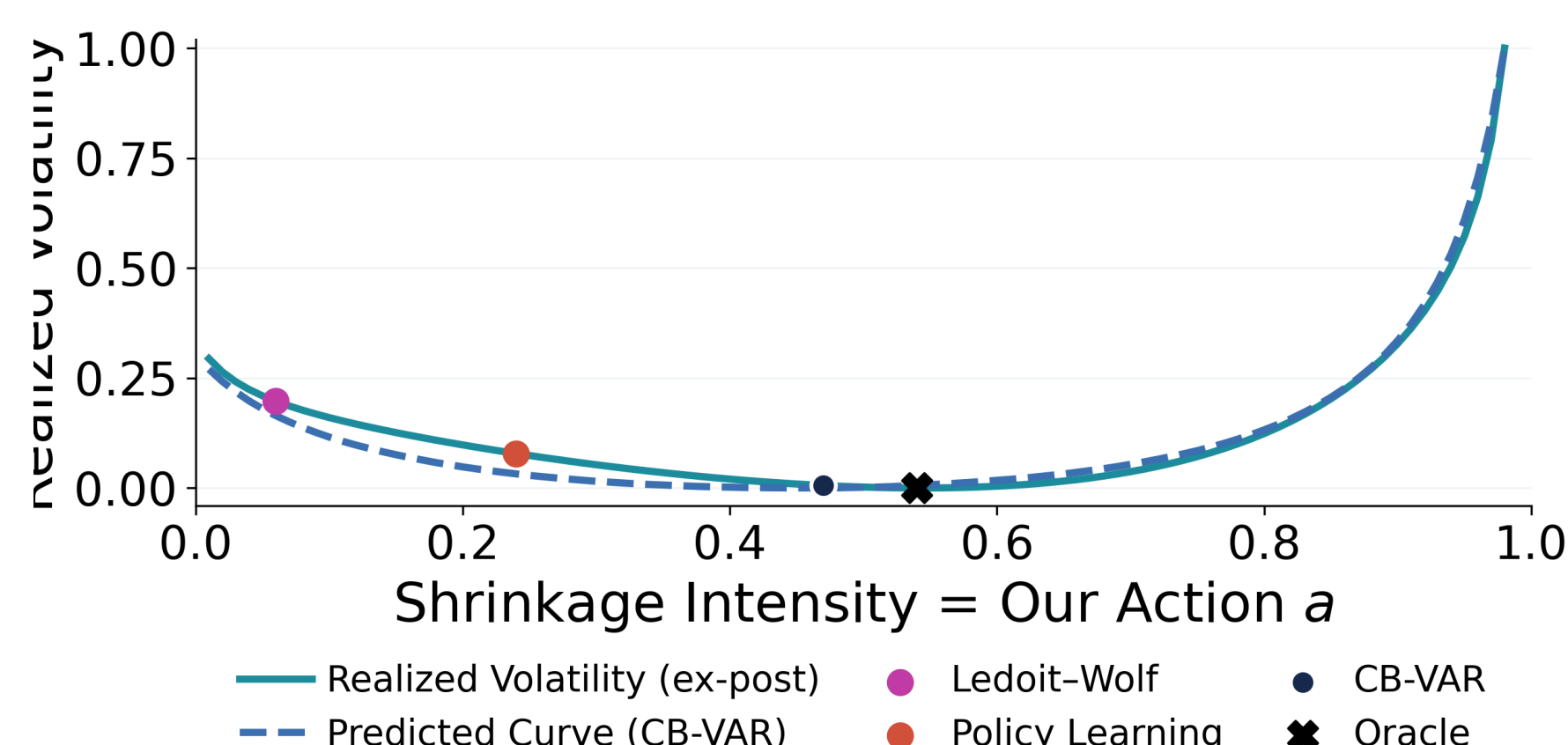
PROBLEM SETUP

- ★ Choose **shrinkage intensity** a to minimize the realized out-of sample portfolio risk.

REGRET

$$\mathcal{R}_t(a_t) = \sqrt{\hat{w}_t(a_t)^\top \Sigma_t^* \hat{w}_t(a_t)} - \sqrt{(w_t^*)^\top \Sigma_t^* w_t^*}$$

- ★ Supervised label a^* is not observable ⇒ **decision-making under uncertainty**.



Example (2001-03-29): our method predicts full reward curve (no lookahead)

- ★ As Σ_t^* changes slowly, we can **assess a ex-post** by portfolio volatility.

REWARD PROXY

$$r(s_t, a_t) = -\sigma(a_t)$$

WHY NOT CROSS-VAL?

- ★ CV assumes that **one global shrinkage** generalizes across multiple folds ⇒ optimum is **regime-independent**.
- ★ However, a_t^* is driven by the economy ⇒ learn a **forward-looking map** $O_t \rightarrow a_t$

WHY NOT POLICY LEARNING?

If discretization error $\equiv 0$, the Regret splits as:

$$C(\hat{\pi}_m) - C(\pi_{\text{oracle}}) = \underbrace{C(\pi_m^*) - C(\pi_{\text{oracle}})}_{\text{risk mismatch}} + \underbrace{C(\hat{\pi}_m) - C(\pi_m^*)}_{\text{estimation error}}$$

POLICY LEARNING

$$C(\hat{\pi}_{\text{PL}}) - C(\pi_{\text{oracle}}) = \underbrace{C(\pi_{\text{PL}}^*) - C(\pi_{\text{oracle}})}_{>0} + \underbrace{C(\hat{\pi}_{\text{PL}}) - C(\pi_{\text{PL}}^*)}_{\text{estimation error}}$$

Risk mismatch survives even as $n \rightarrow \infty$.

CONTEXTUAL BANDITS

$$C(\hat{\pi}_{\text{CB}}) - C(\pi_{\text{oracle}}) = \underbrace{0}_{\text{risk mismatch}} + \underbrace{C(\hat{\pi}_{\text{CB}}) - C(\pi_{\text{CB}}^*)}_{\text{estimation error}}$$

CONTEXTUAL BANDITS

- ★ Each step we transition to a **new state** $s_t \sim p(\cdot | O_t, U_t)$, similar to some past states.
- ★ We observe a **noisy version** of this state $O_t \sim \mu(\cdot | s_t)$.
- ★ Policy acts greedily w.r.t. the **learned reward**.

WHY RL → CB FOR THIS TASK?

$$a_t^* = \pi^*(s_t) = \arg \max_{a \in \mathcal{A}} \left[r(s_t, a) + \gamma \underbrace{\sum_{s_{t+1} \in \mathcal{S}} P(s_{t+1} | s_t, a) V^*(s_{t+1})}_{\text{independent of } a} \right]$$
$$= \arg \max_{a \in \mathcal{A}} r(s_t, a) \Rightarrow \text{Contextual Bandits (one-step)}$$

- ★ The estimator is **Bayes-optimal**.
- ★ **Delayed feedback** — last reward curve is lagged by L steps.



ESTIMATION PIPELINE

Estimate $\hat{a}_t$ from $(X_u, R_u)_{t-H \leq u < t}$ dataset:

1 • Collect Reward Curves $Y_{u,a}$

Backtest each shrinkage intensity; reward is negative realized L -day volatility.

$$\hat{\Sigma} = \Sigma(a, R_{[u-W+1]:[u]}), \quad w(\hat{\Sigma}) \in \mathbb{R}^N,$$

$$Y_{u,a} := r(s_u, a) = \text{Backtest}(w(\hat{\Sigma}), R_{[u+1]:[u+L]})$$

FULL OFFLINE COVERAGE / NO EVALUATION COST ⇒ NO EXPLORATION NEEDED

2 • Encode Curve by RFF & Extract Latent State

Same $\Phi \in \mathbb{R}^{|\mathcal{A}| \times p}$ for every $u \Rightarrow$ betas encode the reward curve shape.

$$\text{Reward Curve: } Y_u(a) \sim \mathcal{GP}(\mu, k) \xrightarrow{\text{RFF}} Y_u(a) \stackrel{\text{Ridge}}{\approx} \phi(a)^\top \hat{\beta}_u$$

$$\text{Latent State: } \Psi_u := [\hat{\beta}_u^\top \ X_u] \in \mathbb{R}^{p+d}$$

FIXED DICTIONARY ⇒ ONLY $\hat{\beta}_i$ VARIES OVER REGIMES

3 • Fit VAR & Roll Forward ($L = 21$)

$f_\theta = \text{VAR}(1)$ propagates the **latent state** through L steps.

For $i = 1, \dots, L$:

$$\hat{\beta}_{t-L+i} = f_\theta(\Psi_{t-L+i-1}^\top)$$

$$\hat{\Psi}_{t-L+i} = [\hat{\beta}_{t-L+i}^\top \ X_{t-L+i}]$$

AUTOREGRESSIVE REGIMES & L -STEPS DELAY OF THE LAST REWARD CURVE

4 • Select Shrinkage a^*

Reconstruct the predicted reward curve and pick the maximizing point.

$$\hat{\beta}_t \in \mathbb{R}^p \Rightarrow \hat{Y}_t = \Phi \hat{\beta}_t \in \mathbb{R}^{|\mathcal{A}|}$$

$$\hat{a}_t = \arg \max_{a \in \mathcal{A}_d} \hat{Y}_{t,a}$$

ARGMAX ON THE LEARNED REWARD → GREEDY W.R.T. $Q^*(s_t, a_t) \Rightarrow$ BAYES-OPTIMAL

STAY TUNED



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KEY RESULT ($GMV, N = 500$)

31.69%

of the policy learning → oracle gap closed

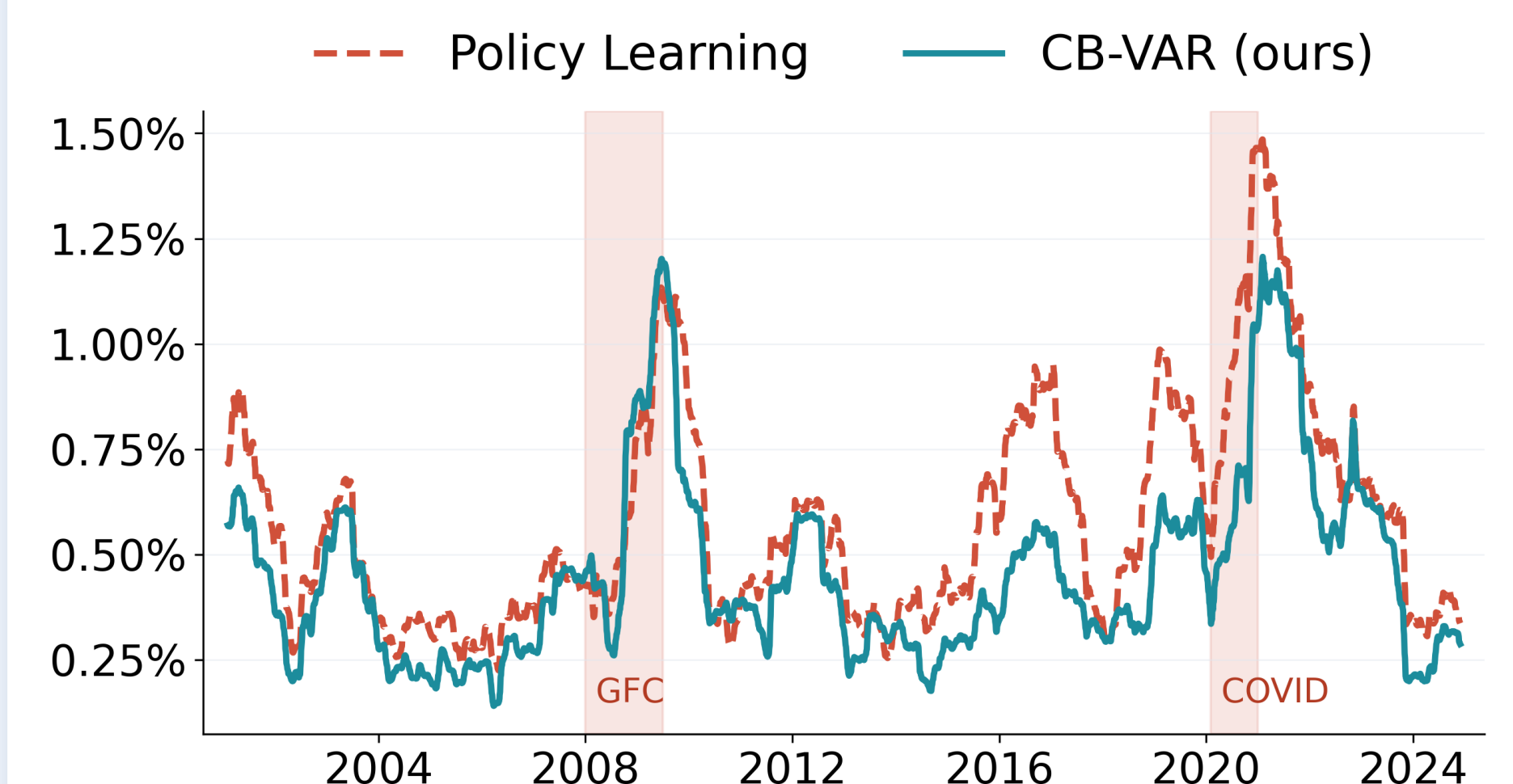
99% confidence interval

[18.01%, 43.24%]

gap of realized annualized volatility · block (42 days) bootstrap

EMPIRICAL RESULTS

Regret vs oracle (annual vol, 1Y moving avg)



Cumulative regret saved vs policy learning



Realized annualized volatility (%), GMV

N	LW	CV	PL	CB-VAR	Oracle	Gap closed
100	13.37	12.85	12.82	12.55	11.82	27.31%
500	11.69	11.31	11.21	10.97	10.46	31.69%
1000	10.16	10.16	10.20	10.07	9.77	30.03%
2000	10.07	10.15	10.16	10.06	9.78	26.85%

Gap closed = share of the PL → oracle gap removed by CB-VAR.
99% bootstrap CIs exclude 0 for all N.

CONCLUSION

- ★ Covariance matrix shrinkage via investor's **economic loss**, not a statistical proxy.
- ★ A **regime-dependent shrinkage** intensity targets OOS portfolio risk.
- ★ The proposed estimator is **Bayes-optimal** w.r.t. the observed features O_t .
- ★ CB-VAR achieves up to 32% **oracle gap reduction**, attains **lower portfolio volatility**.